IS-2 MOTHERBOARD: FOUR ABZ + TWO FREQUENCY + MPG A/B RECEIVER REGION r0.1

U1: four used receiver paths; G and /G tied IO_3V3
0.1uF bypass close to VCC/GND; 1uF local bulk.

U2: four used receiver paths; G and /G tied IO_3V3
0.1uF bypass close to VCC/GND; 1uF local bulk.

U3: four used receiver paths; G and /G tied IO_3V3
0.1uF bypass close to VCC/GND; 1uF local bulk.

U4: four used receiver paths; G and /G tied IO_3V3
0.1uF bypass close to VCC/GND; 1uF local bulk.

C1 100nF 100V X7R
C2 16V X7R
C3 100nF 100V X7R
C4 16V X7R
C5 100nF 100V X7R
C6 16V X7R
C7 100nF 100V X7R
C8 16V X7R

B1 XE0 Encoders / NONPHYSICAL
ENC0_5V_LIMITED
LOGIC_GND
ENC0_A_P
ENC0_A_N
ENC0_B_P
ENC0_B_N
ENC0_Z_P
ENC0_Z_N
LOGIC_GND
CHASSIS

B2 XE1 Encoders / NONPHYSICAL
ENC1_5V_LIMITED
LOGIC_GND
ENC1_A_P
ENC1_A_N
ENC1_B_P
ENC1_B_N
ENC1_Z_P
ENC1_Z_N
LOGIC_GND
CHASSIS

B3 XE2 Encoders / NONPHYSICAL
ENC2_5V_LIMITED
LOGIC_GND
ENC2_A_P
ENC2_A_N
ENC2_B_P
ENC2_B_N
ENC2_Z_P
ENC2_Z_N
LOGIC_GND
CHASSIS

B4 XE3 Encoders / NONPHYSICAL
ENC3_5V_LIMITED
LOGIC_GND
ENC3_A_P
ENC3_A_N
ENC3_B_P
ENC3_B_N
ENC3_Z_P
ENC3_Z_N
LOGIC_GND
CHASSIS

U1 AM26LV32EIPWR
U2 AM26LV32EIPWR
U3 AM26LV32EIPWR
U4 AM26LV32EIPWR
IO_3V3
LOGIC_GND

ENC0_A_FPGA
ENC0_B_FPGA
ENC0_Z_FPGA
ENC1_A_FPGA
ENC1_B_FPGA
ENC1_Z_FPGA
ENC2_A_FPGA
ENC2_B_FPGA
ENC2_Z_FPGA
ENC3_A_FPGA
ENC3_B_FPGA
ENC3_Z_FPGA
FREQ0_FPGA
FREQ1_FPGA
MPG_A_FPGA
MPG_B_FPGA

B5 XF Frequency / NONPHYSICAL
FREQ_5V_LIMITED
LOGIC_GND
FREQ0_P
FREQ0_N
FREQ1_P
FREQ1_N
FREQ_MOD_ID_TO_FRONTEND
FREQ_MOD_PRESENT_TO_FRONTEND
LOGIC_GND
CHASSIS

B6 XH1 MPG / NONPHYSICAL
MPG_5V_LIMITED
LOGIC_GND
MPG_A_P
MPG_A_N
MPG_B_P
MPG_B_N
MPG_VREF_2V5
MPG_ENABLE_TO_STATUS
LOGIC_GND
CHASSIS

R1 120R 0.5% 0.5W / FIT
R2 120R 0.5% 0.5W / FIT
R3 120R 0.5% 0.5W / FIT
R4 120R 0.5% 0.5W / FIT
R5 120R 0.5% 0.5W / FIT
R6 120R 0.5% 0.5W / FIT
R7 120R 0.5% 0.5W / FIT
R8 120R 0.5% 0.5W / FIT
R9 120R 0.5% 0.5W / FIT
R10 120R 0.5% 0.5W / FIT
R11 120R 0.5% 0.5W / FIT
R12 120R 0.5% 0.5W / FIT
R13 120R 0.5% 0.5W / FIT
R14 120R 0.5% 0.5W / FIT
R15 120R 0.5% 0.5W / DNP
R16 120R 0.5% 0.5W / DNP

ENC0_A_P
ENC0_A_N
ENC0_B_P
ENC0_B_N
ENC0_Z_P
ENC0_Z_N
ENC1_A_P
ENC1_A_N
ENC1_B_P
ENC1_B_N
ENC1_Z_P
ENC1_Z_N
ENC2_A_P
ENC2_A_N
ENC2_B_P
ENC2_B_N
ENC2_Z_P
ENC2_Z_N
ENC3_A_P
ENC3_A_N
ENC3_B_P
ENC3_B_N
ENC3_Z_P
ENC3_Z_N
FREQ0_P
FREQ0_N
FREQ1_P
FREQ1_N
MPG_A_P
MPG_A_N
MPG_B_P
MPG_B_N

All 16 receiver channels used; no receiver output NC. No galvanic isolation or cable-line-fault detector here.
14 fitted 120R terminations; MPG R15/R16 DNP to avoid loading a 1mA TTL reference. Differential MPG needs a qualified termination variant.
Terminated/disconnected or shorted pairs do not have guaranteed fail-safe state. Encoder validity must not be inverted from a static output.
No probes, pendant shift register, sensor power limiter, 2.5V reference, accessory identity/presence or status frontend implemented in this receiver region.

B7 EXTERNALLY PROVIDED FEEDS / NONPHYSICAL
ENC0_5V_LIMITED
ENC1_5V_LIMITED
ENC2_5V_LIMITED
ENC3_5V_LIMITED
FREQ_5V_LIMITED
MPG_5V_LIMITED
MPG_VREF_2V5
LOGIC_GND
CHASSIS

B8 FPGA LOGIC BOUNDARY / NONPHYSICAL
IO_3V3
LOGIC_GND
ENC0_A_FPGA
ENC0_B_FPGA
ENC0_Z_FPGA
ENC1_A_FPGA
ENC1_B_FPGA
ENC1_Z_FPGA
ENC2_A_FPGA
ENC2_B_FPGA
ENC2_Z_FPGA
ENC3_A_FPGA
ENC3_B_FPGA
ENC3_Z_FPGA
FREQ0_FPGA
FREQ1_FPGA
MPG_A_FPGA
MPG_B_FPGA

B9 MISSING FRONTEND BREAKOUT / NONPHYSICAL
FREQ_MOD_ID_TO_FRONTEND
FREQ_MOD_PRESENT_TO_FRONTEND
MPG_ENABLE_TO_STATUS

#FLG01 EXTERNALLY FED LOGIC FEED
IO_3V3
#FLG02 EXTERNALLY FED LOGIC FEED
LOGIC_GND

BOUNDARIES ARE NONPHYSICAL: no unverified connector MPN/footprint. Preserve XE0-3 / XF / XH1 logical field ordering.
5V protected feeds and MPG_VREF enter from future parent regions; this block does not generate or protect them.
Input bound: RS422 5.5V maximum differential for termination arithmetic; receiver common-mode −7…+7V, not a field-voltage rating.
External line surge protection, source/cable load, footprints/stencil, thermal/EMC and rates still require qualification. Manufacturing blocked.

16 receiver paths; partial motherboard region: not a manufacturing release

Sheet: /
File: motherboard_fast_inputs.kicad_sch

Title: IS-2 motherboard differential fast inputs

Size: A1	Date: 2026-09-30	**Rev: r0.1**
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